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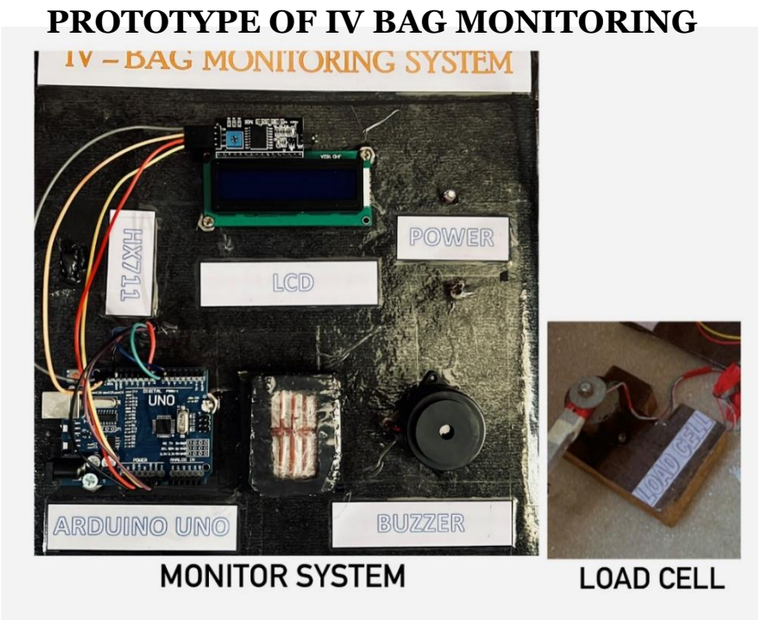
IV – BAG MONITORING SYSTEM

INTRODUCTION:

Our project focuses on the use of intravenous (IV) bags in healthcare management. We recognized the importance of IV bags in various healthcare management and have explored their complexities and challenges. Our project particularly focuses on IV bag monitoring to enhance patient safety and care. Utilizing modern-day technology such as sensors and circuits to address the issue of reverse blood flow in IV bags. We implemented our best practices and technologies, we aim to improve patient safety in IV fluid administration and reducing the ill causes like reverse blood flow.

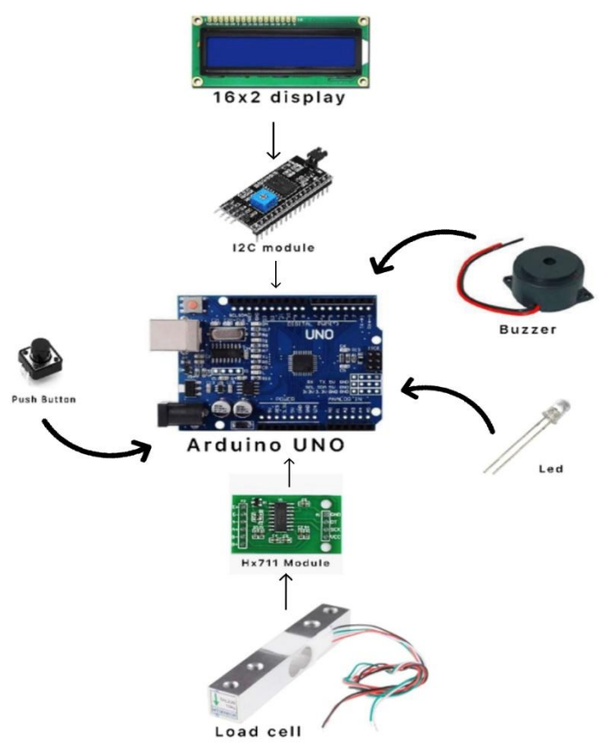
PROTOTYPE OF IV BAG MONITORING



METHODOLOGY:

Our research involves analyzing the existing research, tools and devices used in IV bag monitoring. We made the analysis to identify the gaps, limitations and opportunities for our improvement. We have selected appropriate hardware (load cell to measure) and software components (Arduino UNO is coded for monitoring the IV Bag),for data processing and output. We build a prototype for testing and troubleshooting. See Fig.1 for schematic diagram of the IV bag monitor.

FIG.1



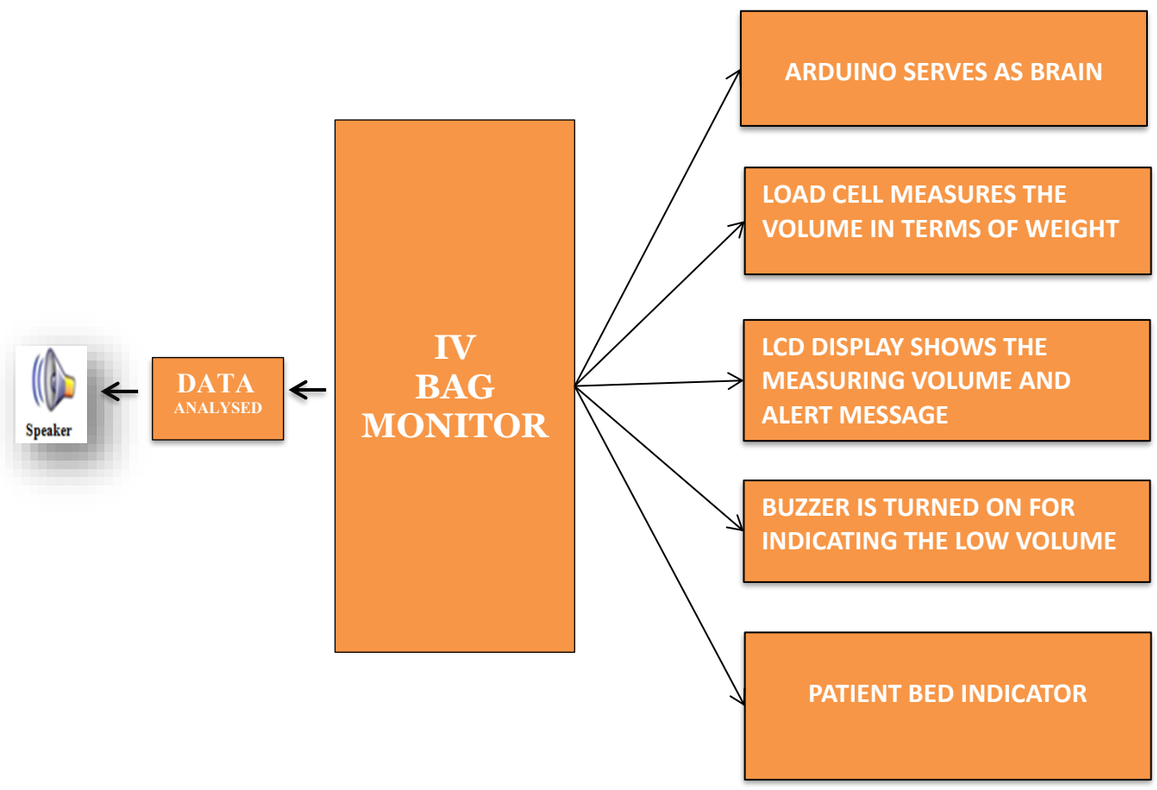
ADVANTAGES

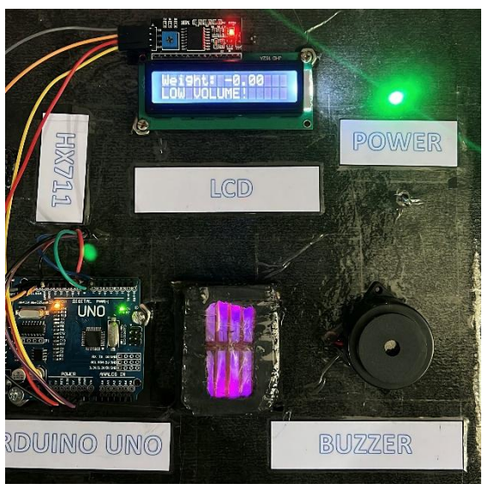
- EARLY DETECTION OF COMPLICATION
- ENHANCED PATIENT SAFETY
- CUSTOMISED ALERT
- COST EFFECTIVENESS
- REAL TIME MONITORING

NEED FOR THE USE OF IV - BAG MONITORING.

- Early Detection: IV bag monitoring systems can detect reverse blood flow in IV bags in real-time, allowing for early detection of this critical issue.
- Enhanced Patient Safety: Reverse blood flow in IV bags can lead to complications such as contamination, clotting, or incorrect dosing.
- IV bag monitoring systems can provide accurate volume of the IV Bag.
- Makes a preventive move towards patient safety.

SPECIFICATION OF THE DESIGN





TEAM MEMBERS:

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